# COUNTERS

## SEQUENTIAL CIRCUITS

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# TOPIC OUTLINE

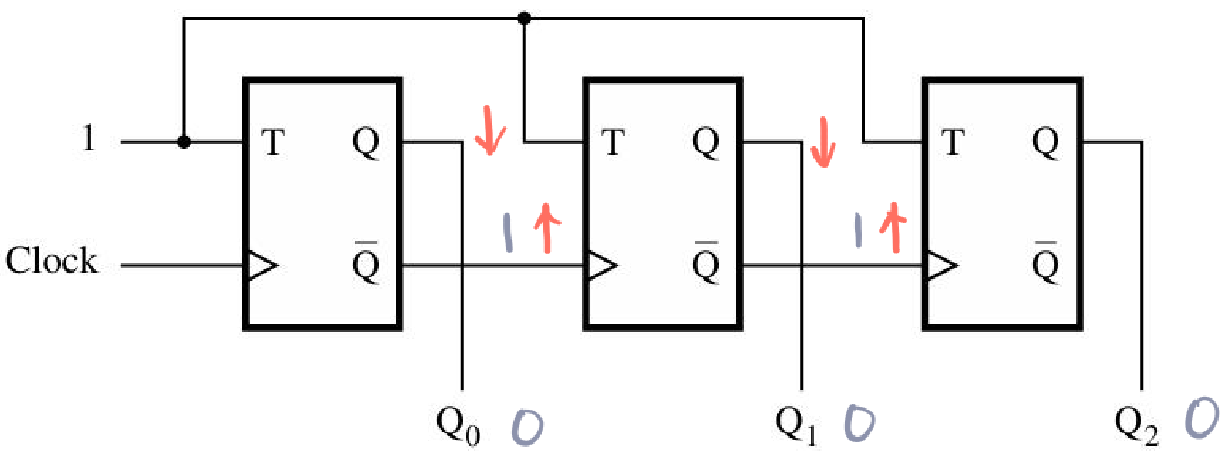
Asynchronous Counter

Synchronous Counter

# ASYNCHRONOUS COUNTERS

# UP-COUNTER

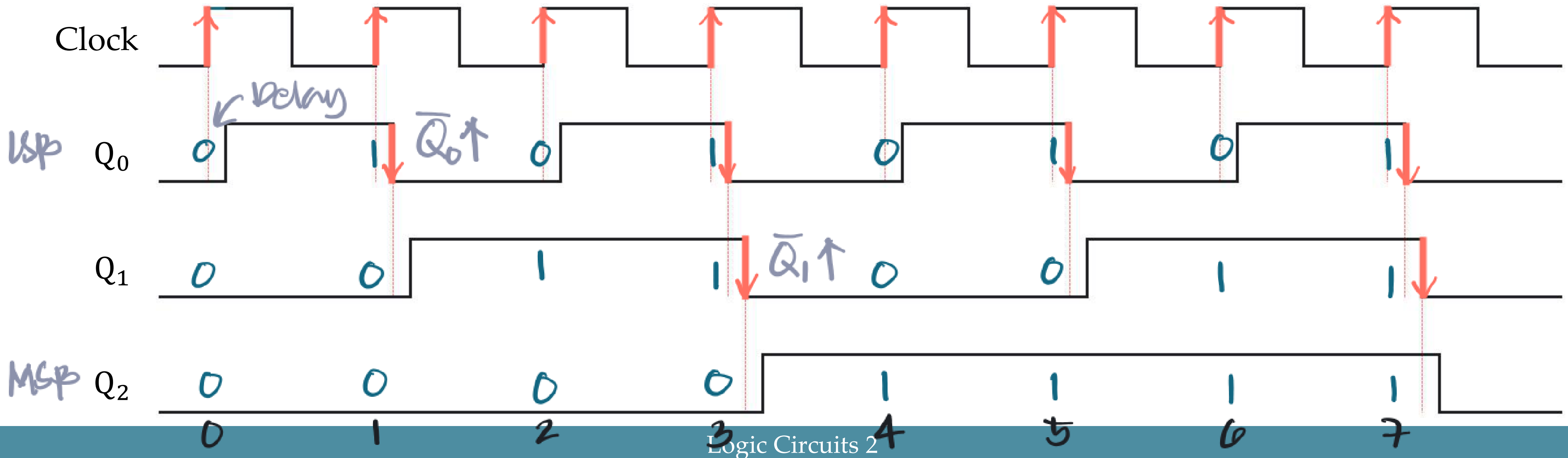
A Three-Bit Up-Counter with T Flip-Flops




T F/F

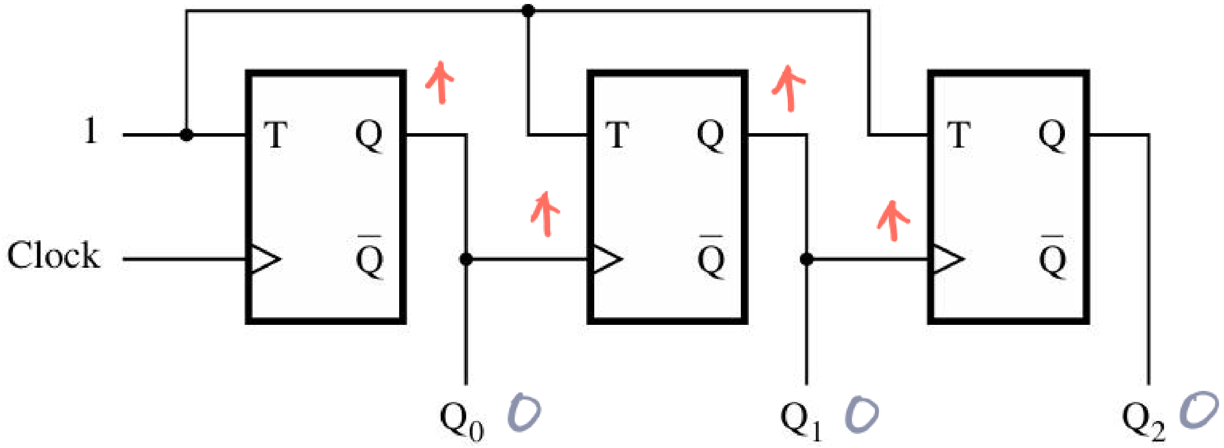
if T=1 then toggle Q

Timing Diagram

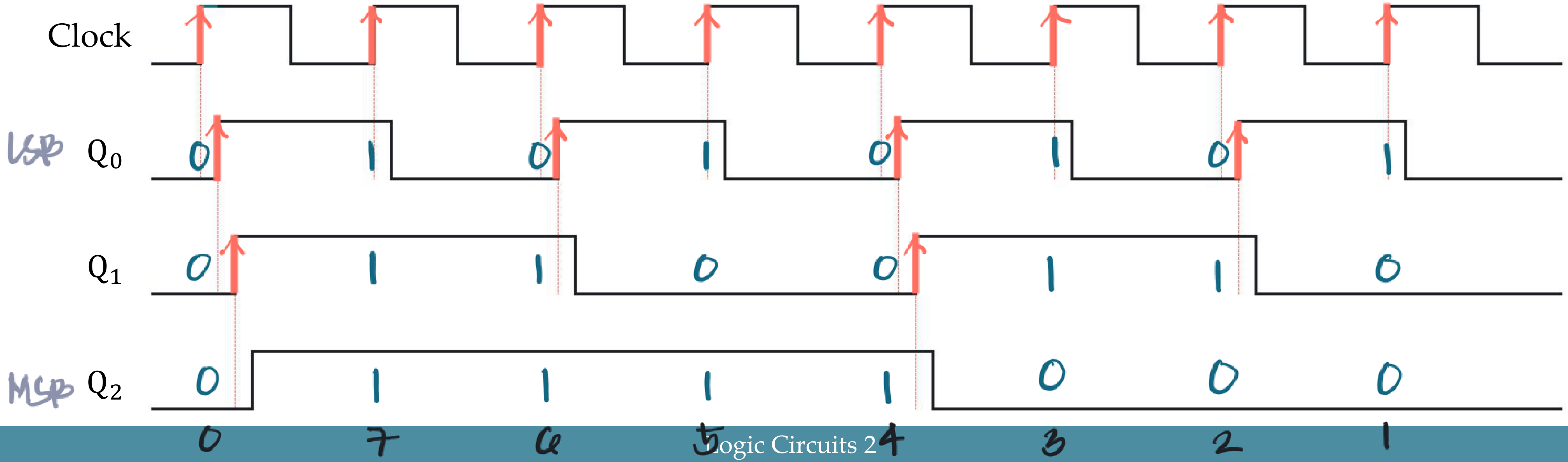


# DOWN-COUNTER

A Three-Bit Down-Counter with T Flip-Flops




Timing Diagram

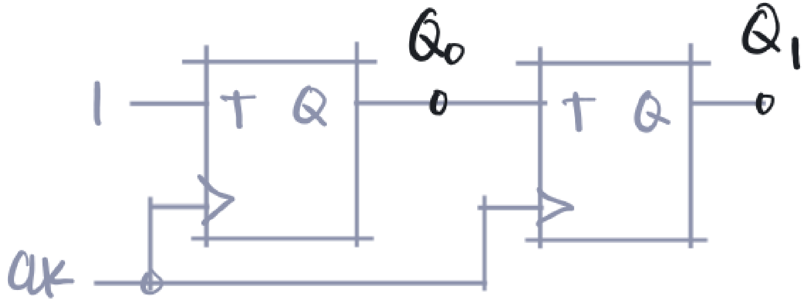


# SYNCHRONOUS COUNTERS

# SYNCHRONOUS COUNTERS

The previous counters are asynchronous – they are simple, but not very fast.

We can build a faster counter by clocking all flip-flops at the same time.




Derivation of the synchronous up-counter

| Clock cycle | $Q_2\,Q_1\,Q_0$ |
|---|---|
| 0 | 0 0 0 |
| 1 | 0 0 1 |
| 2 | 0 1 0 |
| 3 | 0 1 1 |
| 4 | 1 0 0 |
| 5 | 1 0 1 |
| 6 | 1 1 0 |
| 7 | 1 1 1 |
| 8 | 0 0 0 |

$Q_0$ toggles every clk

$Q_1$ toggles when $Q_0 = 1$

# SYNCHRONOUS COUNTERS

Derivation of the synchronous up-counter

| Clock cycle | $Q_2\,Q_1\,Q_0$ |
|---|---|
| 0 | 0 0 0 |
| 1 | 0 0 1 |
| 2 | 0 1 0 |
| 3 | 0 1 1 AND |
| 4 | 1 0 0 |
| 5 | 1 0 1 |
| 6 | 1 1 0 |
| 7 | 1 1 1 AND |
| 8 | 0 0 0 |

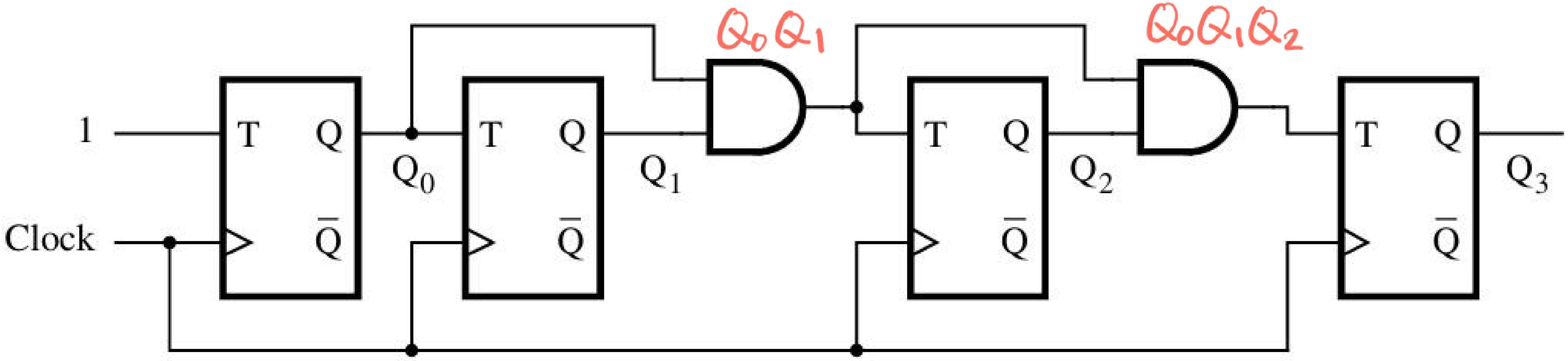
$Q_2$ toggles when both $Q_0 = 1$, $Q_1 = 1$

The previous counters are asynchronous – they are simple, but not very fast.

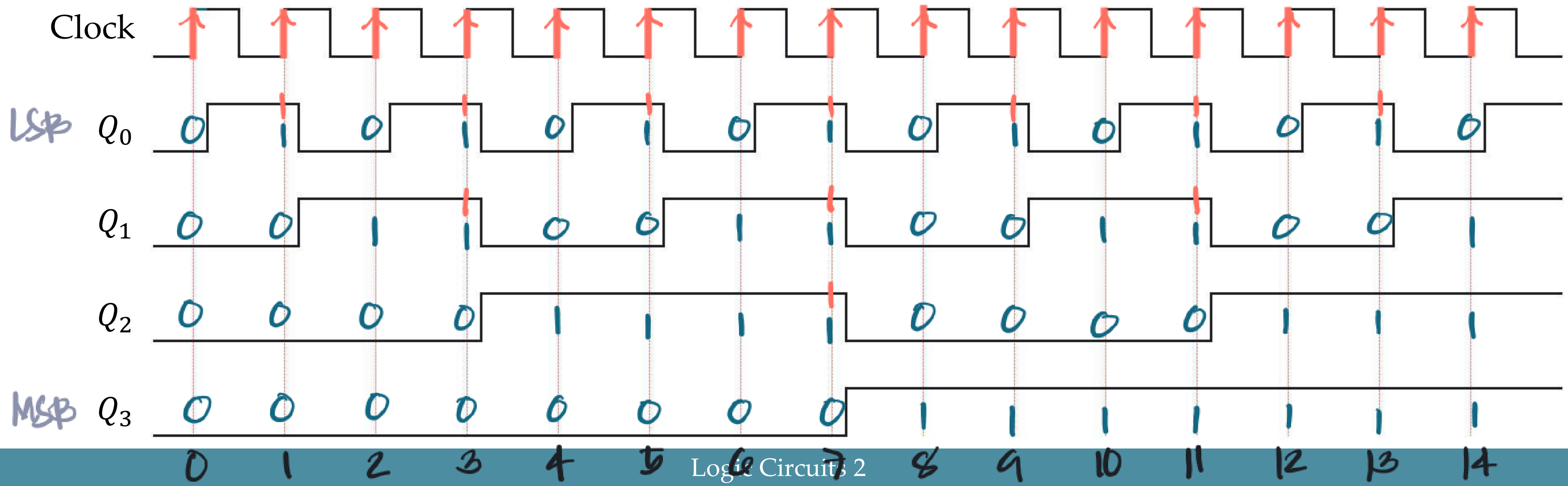
We can build a faster counter by clocking all flip-flops at the **same time**.

# SYNCHRONOUS UP-COUNTER

A four-bit synchronous up-counter with T Flip-Flops

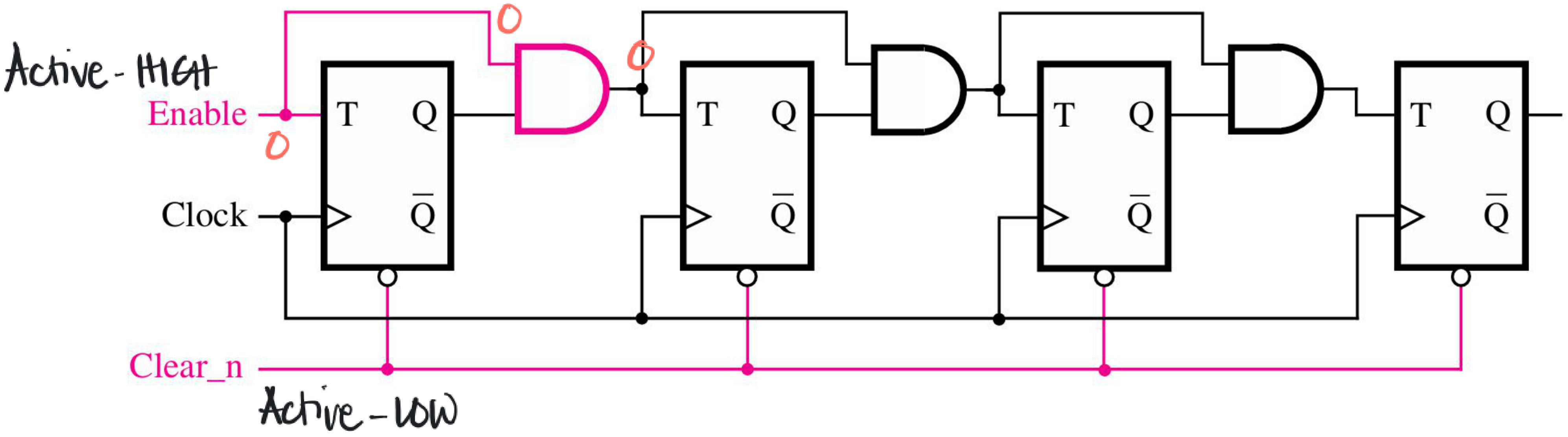



Timing Diagram



# ENABLE AND CLEAR

Inclusion of Enable and Clear capability



# LABORATORY